

COURSE TITLE : VISUAL PROGRAMMING
COURSE CODE : 5254
SEMESTER : 5
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 65
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	INTRODUCTION TO VISUAL PROGRAMMING	15
2	CONTROL STRUCTURES & ARRAYS	15
3	WINDOWS PROGRAMMING	20
4	ADO.NET & ASP.NET	15

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand .NET framework
	2	To know about variables, constants, operators & expressions
2	1	To understand control structures
	2	To know about Arrays
3	1	To understand the concept of windows programming
	2	To know events and exceptions
4	1	To understand ADO.NET
	2	To know about ASP.NET

SPECIFIC OUTCOMES

1.0 Understand Visual Programming

1.1 Explain Visual Programming

- 1.1.1 Define Visual Programming
- 1.1.2 List advantages and disadvantages of visual programming
- 1.1.3 Familiarize .NET Framework
- 1.1.4 Explain .NET Framework architecture
- 1.1.5 Understand CLR and .Net Class Library
- 1.1.6 Compare .NET Languages VB, C#

1.2 Understand Variables, Constants, Operators & Expressions

- 1.2.1 Explain Variables with examples.
- 1.2.2 Explain Constants with examples.

- 1.2.3 Explain Data types and Declaration
- 1.2.4 Explain Arithmetic Operators
- 1.2.5 Explain Relational Operators
- 1.2.6 Explain Logical Operators
- 1.2.7 Explain Operator Precedence
- 1.2.8 Explain Expressions
- 1.2.9 Explain managed code

2.0 Understand Control Structures & Arrays

2.1 Explain Decision and Looping Statements

- 2.1.1 Define If – Then, If – Then - Else Statements. Give examples.
- 2.1.2 Define Select Case Statement. Give examples
- 2.1.3 Define Do While – Loop Statement. Give examples
- 2.1.4 Define For – Next Statement. Give examples.
- 2.1.5 Define Exit and Continue statement. Give examples
- 2.1.6 Define break statement. Give examples

2.2 Familiarize Arrays

- 2.2.1 Explain Single Dimensional Arrays
- 2.2.2 Explain Multi Dimensional Arrays
- 2.2.3 Explain Object Arrays

3.0 Understand Windows Programming

3.1 Familiarize Forms & Controls

- 3.1.1 Describe the procedure of creating Window Forms
- 3.1.2 Explain Form Attributes
- 3.1.3 Define TextBox
- 3.1.4 Define Label
- 3.1.5 Define ComboBox and ListBox
- 3.1.6 Define CheckBox and RadioButton
- 3.1.7 Define Timer
- 3.1.8 Explain Menu Creation
- 3.1.9 Define Dialog Boxes – MsgBox, InputBox

3.2 Understand Events and Exceptions

- 3.2.1 Explain Event Driven Programming
- 3.2.2 Explain Form Events – Load, Activate, Close
- 3.2.3 Explain Keyboard Events – KeyPress, KeyDown
- 3.2.4 Explain Mouse Events – Click, DoubleClick, MouseMove

- 3.2.5 Explain Exception Handling
- 3.2.6 Compare structured and unstructured exception handling.
- 3.2.7 Explain Try – Catch Statement

3.3 Familiarize Reports

- 3.3.1 Explain Crystal Reports
- 3.3.2 Define Database Fields
- 3.3.3 Define Special Fields
- 3.3.4 Define Summary Fields
- 3.3.5 Define Lines & Boxes
- 3.3.6 Define Report Header and Report Footer
- 3.3.7 Define Page Header and Page Footer
- 3.3.8 Define Group Header and Group Footer
- 3.3.9 Define Formatting Fields
- 3.3.10 Define Printer and Page Settings
- 3.3.11 Describe how to create crystal reports
- 3.3.12 List advantages of crystal report

4.0 Understand ADO .NET and ASP .NET

4.1 Introduction to ADO .NET

- 4.1.1 Understand ADO.NET
- 4.1.2 Describe the Architecture of ADO.NET
- 4.1.3 Define Connection Object
- 4.1.4 Define Command Object
- 4.1.5 Define DataAdapter
- 4.1.6 Define DataSet
- 4.1.7 Define DataTable
- 4.1.8 Define DataRow
- 4.1.9 List advantages of ADO.NET

4.2 Know about ASP .NET

- 4.2.1 Explain ASP .NET
- 4.2.2 Design Web Pages using ASP .NET
- 4.2.3 List ASP .NET Controls
- 4.2.4 Explain Web Forms

CONTENT DETAILS

MODULE I: Visual Programming - .NET Framework – CLR - .NET Class Library - .NET Languages – Variables – Constants – Data Types – Operators – Expressions

MODULE II: Control Statements: If Then – If Then Else – Select Case – Looping
Statements: Do While – For – Exit – Continue – Arrays – Single Dimensional –
Multidimensional – Object Arrays

MODULE III: Window Forms - Form Attributes – TextBox – Label – Combo Box – List Box –
Check Box – Radio Button – Timer – Menu - Dialog Boxes – MsgBox –
Input Box - Events – Form Events – Keyboard Events – Mouse Events –
Exceptions – Try Catch Crystal Reports – Database Fields – Special Fields –
Summary Fields – Lines – Boxes - Report Header – Report Footer – Page
Header – Page Footer – Group Header – Group Header – Formatting Fields –
Printer Settings – Page Settings

MODULE IV: ADO .NET – Architecture – Connection – Command – Data Adapter –
DataSet – DataTable – DataRow - ASP .NET – Forms – Controls

BOOKS RECOMMENDED

1. Introduction to Visual basic.NET NIIT Prentice Hall of India
2. Introducing Microsoft .NET David S. Platt Microsoft Press
3. Introduction to Microsoft ASP.NET Work Book, Microsoft Press
4. Danny Ryan & Tommy Ryan, ASP.NET—Wiley Publishers